PYL1002: Waves and Oscillations | **Tutorial - 8**

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IIT Delhi, Spring 2026

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Standing waves, Boundary Conditions, and Fourier Series

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1. Standing Waves

Given:

$$L = 1.2\,\text{m}, \quad \mu = 1.6\,\text{g/cm} = 0.16\,\text{kg/m}, \quad f = 120\,\text{Hz}$$

The tension in the string is provided by a hanging mass:

$$T = mg$$

Wave speed:

$$v = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{mg}{\mu}}$$

As we can see from the figure :

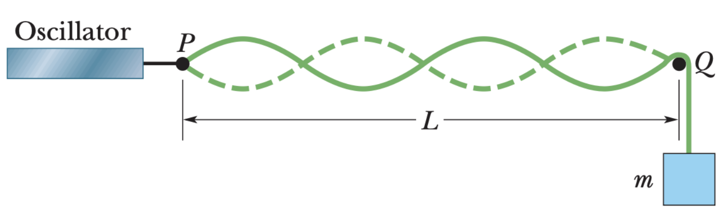
$$2L = n\lambda$$

or,

$$2L = \frac{nv}{f_n}$$

hence allowed frequencies for a string fixed at both ends:

$$\boxed{f_n = \frac{nv}{2L}}$$



(1) Mass for fourth harmonic

For the fourth harmonic ($n = 4$):

$$f = \frac{4v}{2L} = \frac{2v}{L}$$

Substitute v:

$$120 = \frac{2}{L}\sqrt{\frac{mg}{\mu}}$$

Rearranging:

$$\sqrt{\frac{mg}{\mu}} = \frac{120L}{2}$$

$$= \frac{120 \times 1.2}{2} = 72$$

Squaring:

$$\frac{mg}{\mu} = 72^2 = 5184$$

$$m = \frac{5184\mu}{g}$$

$$= \frac{5184 \times 0.16}{9.8} \approx 84.7\,\mathrm{kg}$$

$$\boxed{m \approx 85\,\mathrm{kg}}$$

(2) For $m = 1\,$kg

Wave vilocity :

$$v = \sqrt{\frac{9.8}{0.16}} \approx 7.82\,\mathrm{m/s}$$

so, frequency :

$$f_n = \frac{nv}{2L} = \frac{n \cdot 7.82}{2.4} \approx 3.26n$$

Set equal to driving frequency:

$$120 = 3.26n$$

$$n \approx 36.8$$

Since n must be an integer:

$$\boxed{\text{No exact standing wave is formed}}$$

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2. Orthogonality of Sinusoids

(1) Case $n \neq m$

Consider:

$$I = \int_0^L \sin\left(\frac{n\pi x}{L}\right) \sin\left(\frac{m\pi x}{L}\right) dx$$

Using identity:

$$2 \sin A \sin B = \cos(A - B) - \cos(A + B)$$

$$I = \frac{1}{2} \int_0^L \left[\cos\left(\frac{(n-m)\pi x}{L}\right) - \cos\left(\frac{(n+m)\pi x}{L}\right)\right] dx$$

Now integrate term-by-term:

$$\int \cos(ax)\, dx = \frac{\sin(ax)}{a}$$

Thus:

$$I = \frac{1}{2} \left[\frac{L}{(n-m)\pi} \sin\left(\frac{(n-m)\pi x}{L}\right) - \frac{L}{(n+m)\pi} \sin\left(\frac{(n+m)\pi x}{L}\right)\right]_0^L$$

At $x = L$:

$$\sin((n-m)\pi) = 0, \quad \sin((n+m)\pi) = 0$$

At $x = 0$:

$$\sin(0) = 0$$

Therefore:

$$\boxed{I = 0}$$

(2) Case $n = m$

$$I = \int_0^L \sin^2\left(\frac{n\pi x}{L}\right) dx$$

Using:

$$\sin^2\theta = \frac{1}{2}(1 - \cos 2\theta)$$

$$I = \frac{1}{2} \int_0^L \left[1 - \cos\left(\frac{2n\pi x}{L}\right)\right] dx$$

Split integral:

$$I = \frac{1}{2} \left[\int_0^L dx - \int_0^L \cos\left(\frac{2n\pi x}{L}\right) dx\right]$$

$$I = \frac{I_1 - I_2}{2}$$

$$I_1 = \int_0^l dx = L$$

$$I_2 = \int_0^L cos(\frac{2n\pi x}{L}) = \frac{2n\pi}{L}[sin(2n\pi) - sin(0)] = 0$$

Hence:

$$I = \frac{I_1}{2} = \frac{1}{2}L$$

$$\boxed{I = \frac{L}{2}}$$

3. Heat Equation

(a) Separation of Variables

Assume:

$$T(x,t) = X(x)G(t)$$

Substitute into:

$$\frac{\partial T}{\partial t} = \alpha\frac{\partial^2 T}{\partial x^2}$$

$$X(x)\dot{G}(t) = \alpha G(t)X''(x)$$

Divide both sides by $T(x,t)$:

$$\frac{\dot{G}}{G} = \alpha\frac{X''}{X}$$

(i)

LHS depends only on t, RHS only on x, hence both should be equal to a constant let:

$$\frac{\dot{G}}{G} = \alpha\frac{X''}{X} = -\alpha\lambda = b(constant)$$

(ii)

since,

$$\frac{\dot{G}}{G} = -a\lambda$$

$$\dot{G} = -a\lambda G$$

$$\frac{\mathrm{d}G}{\mathrm{d}t} = -a\lambda G$$

or,

$$\frac{dG}{G} = -a\lambda dt$$

$$\int_0^t \frac{dG}{G} = \int_0^t -a\lambda dt$$

$$\implies G(t) = e^{-a\lambda t}$$

If $\lambda \leq 0$, exponential grows $\Rightarrow$ unphysical.
Thus:

$$\lambda > 0$$

(iii)

similarly for the spatial part ,we have

$$\frac{X''}{X} = -\lambda$$

or

$$X'' + \lambda X = 0$$

as we can see this is equation of SHM, so
general solution:

$$X(x) = A\sin\left(\sqrt{\lambda}x\right) + B\cos\left(\sqrt{\lambda}x\right)$$

Boundary condition:

$$X(0) = 0 \Rightarrow B = 0$$

$$X(L) = 0 \Rightarrow \sin\left(\sqrt{\lambda}L\right) = 0$$

$$\sqrt{\lambda}L = n\pi$$

$$\lambda_n = \frac{n^2\pi^2}{L^2}$$

so the mode function ,

$$X_n(x) = A\sin\left(\frac{n\pi x}{L}\right)$$

where, A is a constant

(iv)

as we already know ,

$$\dot{G} = -\alpha\lambda_n G$$

$$G_n(t) = e^{-\alpha\lambda_n t} = e^{-t/\tau_n}$$

where

$$\tau_n = \frac{1}{\alpha\lambda_n} = \frac{L^2}{\alpha n^2\pi^2} \quad (proved)$$

(v)

since

$$T(x,t) = X(x)G(t)$$

putting values of$X(x)$ and$G(t)$

$$T(x,t) = \sum_{n=1}^{\infty} C_n \sin\left(\frac{n\pi x}{L}\right) e^{-t/\tau_n}$$

Only one constant per mode because equation is first-order in time.

(b) Fourier Coefficients

We are given the initial temperature profile:

$$T(x,0) = \frac{4T_0}{L^2}\, x(L-x)$$

The Fourier coefficients are:

$$C_n = \frac{2}{L}\int_0^L \frac{4T_0}{L^2} x(L-x)\sin\left(\frac{n\pi x}{L}\right)\, dx$$

$$= \frac{8T_0}{L^3}\int_0^L x(L-x)\sin\left(\frac{n\pi x}{L}\right)\, dx$$

Define the integral:

$$I = \int_0^L x(L-x)\sin\left(\frac{n\pi x}{L}\right)\, dx$$

Expand:

$$x(L-x) = Lx - x^2$$

Thus:

$$I = \int_0^L Lx\sin\left(\frac{n\pi x}{L}\right) dx - \int_0^L x^2\sin\left(\frac{n\pi x}{L}\right) dx$$

Let:

$$I = I_1 - I_2$$

Step 1: Evaluate $I_1 = \int_0^L Lx\sin\left(\frac{n\pi x}{L}\right) dx$

Take $u = x$, $dv = \sin(ax)\, dx$, where $a = \frac{n\pi}{L}$

$$du = dx, \quad v = -\frac{\cos(ax)}{a}$$

$$I_1 = L\left[-\frac{x\cos(ax)}{a}\bigg|_0^L + \frac{1}{a}\int_0^L \cos(ax)\, dx\right]$$

Evaluate:

$$\int_0^L \cos(ax)\,dx = \frac{\sin(aL)}{a} = \frac{\sin(n\pi)}{a} = 0$$

Thus:

$$I_1 = -\frac{L}{a}\left[x\cos(ax)\right]_0^L$$

$$= -\frac{L}{a}\left[L\cos(n\pi) - 0\right]$$

$$I_1 = -\frac{L^2}{a}\cos(n\pi)$$

Step 2: Evaluate $I_2 = \int_0^L x^2 \sin(ax)\,dx$

Use integration by parts twice.

First integration:

$$u = x^2, \quad dv = \sin(ax)\,dx$$

$$du = 2x\,dx, \quad v = -\frac{\cos(ax)}{a}$$

$$I_2 = -\frac{x^2\cos(ax)}{a}\bigg|_0^L + \frac{2}{a}\int_0^L x\cos(ax)\,dx$$

Now evaluate:

$$J = \int_0^L x\cos(ax)\,dx$$

Second integration:

$$u = x, \quad dv = \cos(ax)\,dx$$

$$du = dx, \quad v = \frac{\sin(ax)}{a}$$

$$J = \frac{x\sin(ax)}{a}\bigg|_0^L - \frac{1}{a}\int_0^L \sin(ax)\,dx$$

$$\int_0^L \sin(ax)\,dx = \frac{1-\cos(aL)}{a}$$

Thus:

$$J = \frac{L\sin(n\pi)}{a} - \frac{1}{a^2}(1-\cos(n\pi))$$

$$= -\frac{1}{a^2}(1-\cos(n\pi))$$

Now substitute back:

$$I_2 = -\frac{L^2\cos(n\pi)}{a} + \frac{2}{a}\left[-\frac{1}{a^2}(1-\cos(n\pi))\right]$$

$$I_2 = -\frac{L^2\cos(n\pi)}{a} - \frac{2}{a^3}(1-\cos(n\pi))$$

Combine $I = I_1 - I_2$

$$I = \left(-\frac{L^2}{a}\cos(n\pi)\right) - \left[-\frac{L^2}{a}\cos(n\pi) - \frac{2}{a^3}(1-\cos(n\pi))\right]$$

$$= \frac{2}{a^3}(1-\cos(n\pi))$$

putting value of $a = \frac{n\pi}{L}$

$$a^3 = \frac{n^3\pi^3}{L^3}$$

$$I = \frac{2L^3}{n^3\pi^3}(1-\cos(n\pi))$$

since ,

$$C_n = \frac{8T_0}{L^3}\cdot I$$

or

$$C_n = \frac{8T_0}{L^3}\cdot\frac{2L^3}{n^3\pi^3}(1-\cos(n\pi))$$

$$\boxed{C_n = \frac{16T_0}{n^3\pi^3}(1-\cos(n\pi))}$$

now for n

Recall:

$$\cos(n\pi) = (-1)^n$$

$$1-\cos(n\pi) = \begin{cases} 0 & n \text{ even} \\ 2 & n \text{ odd} \end{cases}$$

Thus:

$$\boxed{C_n = \begin{cases} \frac{32T_0}{n^3\pi^3}, & n \text{ odd} \\ 0, & n \text{ even} \end{cases}}$$

Since, the function $x(L-x)$ is symmetric about $x = L/2$ and even n modes are antisymmetric about $x = L/2$ because , Product of symmetric and antisymmetric functions integrates to zero.

$$\boxed{\text{Therefore, only odd modes contribute}}$$

(c) Numerical Estimates

Given that $L = 4.0m$, $\alpha = 1.12 \times 10^{-6}$, $T_0 = 1200°C$

(i)

$$\tau_1 = \frac{L^2}{\alpha\pi^2} = \frac{16}{1.12 \times 10^{-6}\pi^2} \approx 1.45 \times 10^6 \,\text{s} \approx 403 \,\text{hours}$$

(ii)

since for odd n, $\quad C_n = \frac{32T_0}{n^3\pi^3}$ and, $\quad \tau_n = \frac{L^2}{\alpha\pi^2 n^2}$

$$\frac{C_3}{C_1} = \frac{1}{27}, \quad \tau_3 = \frac{\tau_1}{9}$$

$$\frac{C_3 e^{-t/\tau_3}}{C_1 e^{-t/\tau_1}} = \frac{1}{27}e^{-8} \approx 10^{-4}$$

this tells us that, higher modes decay very fast.

(iii)

$$T(L/2, t) = C_1 e^{-t/\tau_1}$$

$$C_1 = \frac{32T_0}{\pi^3}$$

$$1240e^{-t/\tau_1} = 100$$

$$t = \tau_1 \ln(12.4) \approx 1000 \,\text{hours}$$

This is an underestimate,because model ignores internal heat generation In reality, fuel keeps producing heat i.e cooling is slower.

(d) Conceptual

(i)

Both the heat equation and wave equation have the same spatial eigenfunctions:

$$\sin\left(\frac{n\pi x}{L}\right)$$

However, their temporal parts differ:

- Heat equation has real exponential :

$$G_n(t) = e^{-\lambda_n t}$$

- Wave equation has complex exponential:

$$G_n(t) = e^{\pm i\omega_n t}$$

we can see that $e^{-\lambda_n t}$ decays exponentially to zero as $t \to \infty$ and $e^{\pm i\omega_n t}$ has constant magnitude:

$$|e^{i\omega_n t}| = 1$$

Thus:

The real exponential decay in the heat equation leads to loss of information (irreversibility),whereas the purely oscillatory complex exponential in the wave equation preserves energy and is reversible.

(ii) Schrödinger Equation in a 1D Box

The time-dependent Schrödinger equation is:

$$i\hbar\frac{\partial\Psi}{\partial t} = -\frac{\hbar^2}{2m}\frac{\partial^2\Psi}{\partial x^2}$$

Assume separable solution:

$$\Psi(x,t) = \psi(x)G(t)$$

Substituting and separating variables:

$$\frac{1}{G}\frac{dG}{dt} = -\frac{i}{\hbar}E$$

$$\Rightarrow G(t) = e^{-iEt/\hbar}$$

The spatial equation is identical to the particle-in-a-box problem:

$$\psi_n(x) = \sin\left(\frac{n\pi x}{L}\right)$$

hence , the total wavefunction can be written as,

$$\Psi_n(x,t) = \sin\left(\frac{n\pi x}{L}\right)e^{-iE_n t/\hbar}$$

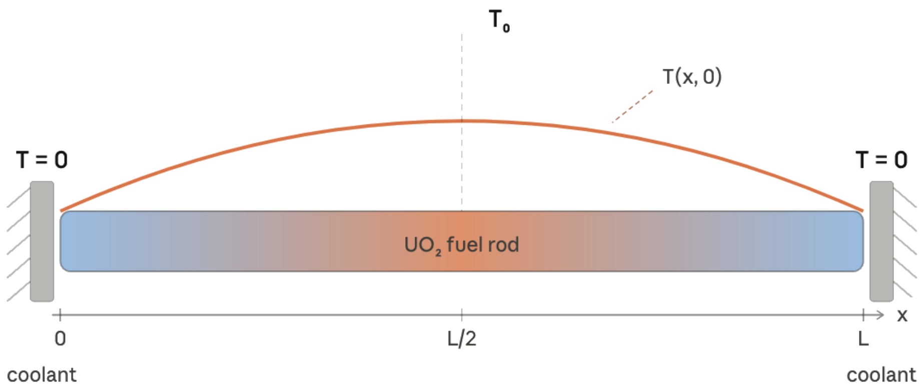
since ,The time factor is purely oscillatory:

$$|G_n(t)| = 1$$

so no decay occurs,

$\rightarrow$ Probability density remains constant in time

$$\boxed{\text{Quantum evolution is reversible because the time dependence is unitary (pure phase).}}$$



4. A Square Wave on a String

Part (a)

To find the equation for a generic normal mode in this system, we start with the generic waveform

$$f_n(x,t) = A_n \sin(k_n x)\cos(\omega_n t) + B_n \sin(k_n x)\sin(\omega_n t) + C_n \cos(k_n x)\cos(\omega_n t) + D_n \cos(k_n x)\sin(\omega_n t). \quad (1)$$

Since the string is pinned at both ends, we know the normal modes must be zero at $x = 0$ and $x = L$. Thus we cannot have any terms with $\cos(k_n x)$ because that would violate the boundary conditions, so we are left with

$$f_n(x,t) = A_n \sin(k_n x)\cos(\omega_n t) + B_n \sin(k_n x)\sin(\omega_n t). \quad (2)$$

Additionally, we know we are releasing the system from rest, so

$$\frac{d}{dt} f_n(x,0) = 0 \Rightarrow B_n \omega_n \sin(k_n x) = 0 \Rightarrow B_n = 0. \quad (3)$$

Thus our generic normal mode must take the form

$$f_n(x,t) = A_n \sin(k_n x)\cos(\omega_n t). \quad (4)$$

Since we can express any wave as the sum of normal modes, we have

$$f(x,t) = \sum_{n=1}^{\infty} A_n \sin(k_n x)\cos(\omega_n t). \quad (5)$$

Now we will plug in our initial conditions. We can express the initial condition as a Fourier sine series, so that

$$f(x,0) = \sum_{n=1}^{\infty} A_n \sin\left(\frac{n\pi}{L} x\right). \quad (6)$$

Note this is a sum of sines, so we can do a Fourier analysis on it as done on p. 56 of the lecture notes. The result is that

$$A_n = \frac{2}{L}\int_0^L \sin\left(\frac{n\pi}{L}x\right) f(x,0)dx. \tag{7}$$

We know

$$f(x,0) = \begin{cases} h & L/3 < x < 2L/3 \\ 0 & \text{otherwise.} \end{cases}$$

So we can rewrite our integral as

$$\begin{aligned} A_n &= \frac{2}{L}\int_{\frac{L}{3}}^{\frac{2L}{3}} h\sin\left(\frac{n\pi}{L}x\right)dx = -\frac{2hL}{Ln\pi}\left(\cos\left(\frac{2n\pi}{3}\right) - \cos\left(\frac{n\pi}{3}\right)\right) \\ &= \frac{2h}{n\pi}\left(\cos\left(\frac{n\pi}{3}\right) - \cos\left(\frac{2n\pi}{3}\right)\right). \end{aligned} \tag{8}$$

Then we have solved for A_n. So, we get our final expression for $f(x,t)$ is

$$\boxed{f(x,t) = \sum_{n=1}^{\infty}\frac{2h}{n\pi}\left(\cos\left(\frac{n\pi}{3}\right) - \cos\left(\frac{2n\pi}{3}\right)\right)\sin\left(\frac{n\pi}{L}x\right)\cos\left(\frac{n\pi v}{L}t\right)} \tag{9}$$

where we've subbed in $\omega_n = \frac{n\pi v}{L}$.

Part (b)

For the wave to return to its initial shape, we solve for

$$f(x,t) = f(x,0). \tag{10}$$

In simpler terms, from (9), we only need to solve for when

$$\cos\left(\frac{n\pi v}{L}t\right) = 1, \quad n \in \mathbb{Z}^+ \tag{11}$$

holds true. (we are interested in the nontrivial solutions, so we ignore the $n = 0$ case.) This means for $\forall n \in \mathbb{Z}^+$, we need some integer m such that

$$\frac{n\pi v}{L}t = 2\pi m. \tag{12}$$

Note that the m's can be different for different n. Rearranging gives our requirement as

$$t = \frac{2L}{v} \times \frac{m}{n}. \tag{13}$$

We must choose an m for each n so that t will be the same for every value of m, which means m/n is a constant. It is easy to see that if we choose $m = n$, then m will be an integer no matter the n, and that $t = 2L/v$. In fact, we can choose $m = 2n$ or $m = 3n$, or m to be any integer multiple of n, and we'll still have m as an integer always and t the same for all n. Thus we **will see** a return to our original shape, and it will happen at

$$\boxed{t = \frac{2L}{v}m, \quad m \in \mathbb{Z}^+.} \tag{14}$$

5. Open Boundary Conditions

We start from the general solution of the wave equation for a string:

$$y(x,t) = (A\cos kx + B\sin kx)(C\cos\omega t + D\sin\omega t)$$

where

$$\omega = vk, \quad v = \sqrt{\frac{T}{\mu}}$$

(1) Allowed values of k for fix-open boundary condition

We are given:

- Left end fixed: $y(0,t) = 0$
- Right end free: no transverse force $\Rightarrow \frac{\partial y}{\partial x} = 0$ at $x = L$

$$y(0,t) = (A\cos 0 + B\sin 0)(C\cos\omega t + D\sin\omega t)$$

$$= A(C\cos\omega t + D\sin\omega t)$$

For this to be zero for all t:

$$A = 0$$

Thus, solution reduces to:

$$y(x,t) = B\sin(kx)(C\cos\omega t + D\sin\omega t)$$

At a free end, the transverse force must vanish. Since force $\propto \frac{\partial y}{\partial x}$:

$$\frac{\partial y}{\partial x} = Bk\cos(kx)(C\cos\omega t + D\sin\omega t)$$

At $x = L$:

$$\left.\frac{\partial y}{\partial x}\right|_{x=L} = Bk\cos(kL)(C\cos\omega t + D\sin\omega t)$$

For this to vanish for all t:

$$\cos(kL) = 0$$

$$\cos(kL) = 0 \Rightarrow kL = \frac{(2n+1)\pi}{2}, \quad n = 0, 1, 2, \ldots$$

Thus:

$$\boxed{k_n = \frac{(2n+1)\pi}{2L}}$$

(2) Allowed frequencies

Using dispersion relation:

$$\omega_n = vk_n = v\frac{(2n+1)\pi}{2L}$$

Frequency:

$$f_n = \frac{\omega_n}{2\pi} = \frac{v}{2\pi} \cdot \frac{(2n+1)\pi}{2L}$$

$$\boxed{f_n = \frac{(2n+1)v}{4L}}$$

Thus only **odd harmonics** exist.

(3) Open-open pipe (Bansuri)

At an open end, the air displacement is free to oscillate, analogous to a free end of a string. Therefore:

$$\frac{\partial y}{\partial x} = 0 \quad \text{at both ends}$$

General solution:

$$y(x,t) = (A\cos kx + B\sin kx)(\text{time part})$$

Apply boundary conditions:

At $x = 0$:

$$\frac{\partial y}{\partial x} = -Ak\sin 0 + Bk\cos 0 = Bk = 0$$

$$\Rightarrow B = 0$$

Thus:

$$y(x,t) = A\cos(kx)(\text{time part})$$

At $x = L$:

$$\frac{\partial y}{\partial x} = -Ak\sin(kL) = 0$$

$$\Rightarrow \sin(kL) = 0$$

$$kL = n\pi$$

Thus:

$$k_n = \frac{n\pi}{L}$$

Frequency:

$$f_n = \frac{vk_n}{2\pi} = \frac{nv}{2L}$$

$$\boxed{f_n = \frac{nv}{2L}}$$

This is identical to the fixed-fixed case.

(4) Comparison: Bansuri vs Shehnai

Bansuri (open-open):

$$f_1 = \frac{v}{2L}$$

Shehnai (closed-open):

$$f_1 = \frac{v}{4L}$$

Thus:

$$\frac{f_{\text{shehnai}}}{f_{\text{bansuri}}} = \frac{v/4L}{v/2L} = \frac{1}{2}$$

$$\boxed{f_{\text{shehnai}} = \frac{1}{2} f_{\text{bansuri}}}$$

Physical interpretation:

- Bansuri supports all harmonics: $f, 2f, 3f, \ldots$
- Shehnai supports only odd harmonics: $f, 3f, 5f, \ldots$
- Fundamental of shehnai is half that of bansuri $\Rightarrow$ lower pitch

$$\boxed{\text{Shehnai produces a deeper (lower frequency) sound than bansuri for the same length}}$$
